CARLA ML Safety Baseline Models

Evaluation & Safety Analysis Report

Introduction to Machine Learning Safety

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ResNet-18 · PyTorch · CARLA Simulator Dataset

1. Project Overview

This report presents the training, evaluation, and safety analysis of three binary perception models developed for the CARLA autonomous driving simulator. The project was completed as part of the Introduction to Machine Learning Safety course and contributes evidence toward a structured safety case for the system.

1.1 System Description

The system consists of three independent binary classifiers, each taking a single forward-facing RGB camera image as input and producing a binary prediction:

- Pedestrian Detector — has_pedestrian
- Traffic Light Detector — has_traffic_light
- Vehicle Detector — has_vehicle

These outputs feed directly into the vehicle autopilot decision logic, making correct predictions safety-critical. All three models share the same ResNet-18 backbone architecture and were trained exclusively on clear-weather daytime data.

1.2 Technologies

- Python 3 with PyTorch 2.12.0
- ResNet-18 pretrained backbone (fine-tuned for binary classification)
- Binary Cross-Entropy (BCE) Loss
- Adam Optimizer
- Pandas, NumPy, Matplotlib
- Jupyter Notebook + Git/GitHub

2. Dataset Exploration (Exercise 3.4)

2.1 Dataset Splits

The CARLA dataset contains front-facing RGB camera images collected in simulated urban driving environments under clear/sunny weather and daytime conditions.

The dataset is split as follows:

Split	Images
Training	7,200
Validation	3,600
Test (Standard)	3,600
Test-Fog	ODD variant
Test-Night	ODD variant
Test-Town-01	ODD variant

2.2 Class Distribution Analysis

The training set reveals significant class imbalance across all three labels:

Label	Positive (True)	Negative (False)	Positive Ratio
has_pedestrian	1,718	5,482	23.9% ⚠ Minority
has_traffic_light	5,276	1,924	73.3% Majority
has_vehicle	5,458	1,742	75.8% Majority

Key observation: The pedestrian label is severely under-represented at only 23.9% positive rate. This imbalance directly causes the poor pedestrian recall seen in evaluation results, as the model learns to predict 'no pedestrian' by default to minimise loss.

3. Model Architecture & Training (Exercise 3.5)

3.1 Architecture

All three classifiers share the same architecture:

- Backbone: ResNet-18 pre-trained on ImageNet
- Final fully-connected layer replaced with a single-neuron output
- Sigmoid activation for binary probability output
- Threshold: 0.5 for binary decision

3.2 Training Configuration

Hyperparameter	Value
Loss Function	Binary Cross-Entropy (BCE)
Optimizer	Adam
Epochs	5
Device	CPU (MacBook Air)
Pre-trained Backbone	Yes — ImageNet weights (transfer learning)

3.3 Training Loss Curves

Model	Epoch 1	Epoch 2	Epoch 3	Epoch 4	Epoch 5
Pedestrian	1.0457	0.9702	0.9159	0.8870	0.8415
Traffic Light	0.1805	0.1006	0.0681	0.0605	0.0499
Vehicle	0.7441	0.6360	0.5533	0.4938	0.4618

Analysis: The Traffic Light model converged excellently (loss approaching 0.05), consistent with its class-balanced dataset and high test scores. The Pedestrian model showed the slowest convergence and highest final loss (0.84), indicating insufficient training epochs and the impact of class imbalance. The Vehicle model converged moderately but did not reach a low plateau, suggesting more epochs would help.

3.4 Why Three Separate Models? (Safety Perspective)

Training three separate binary classifiers is preferable to a single multi-label model for the following safety reasons:

1. Fault isolation: A failure in one classifier does not corrupt predictions of others. In a multi-label model, a bug or distribution shift affecting one output could degrade all outputs simultaneously.
2. Independent safety validation: Each model can be separately validated, tested, and certified to meet its own safety threshold (e.g., minimum recall for pedestrian detection).
3. Independent deployment control: Safety engineers can disable or replace a single classifier (e.g., the night-failed traffic light model) without rebuilding the entire system.
4. Clearer accountability: Each model has a single, well-defined function, making it easier to trace errors and assign responsibility in post-incident analysis.

4. Baseline Evaluation Results (Exercises 3.6 & 4.7)

4.1 Standard Test Set — All Models

Model	Accuracy	Precision	Recall	F1 Score	Status
Traffic Light	0.9206	0.9527	0.9358	0.9442	✓ Excellent
Vehicle	0.7875	0.8617	0.8537	0.8577	~ Good
Pedestrian	0.7058	0.1504	0.1076	0.1255	✗ Critical Failure

4.2 Per-Model Analysis

Traffic Light Detector — Excellent

The traffic light detector achieves near-perfect performance across all metrics. The high F1 of 0.944 is driven by the class-majority distribution (73% positive), which aligns well with standard BCE loss training. The model learned clear visual features (traffic light shapes, colours) reliably reproduced in CARLA's controlled simulator environment.

Vehicle Detector — Good

The vehicle detector performs well with recall of 0.854 and precision of 0.862. Vehicles are large, visually salient objects that appear frequently (76% positive rate), aiding learning. The final training loss of 0.46 suggests the model had not fully converged and may improve with more epochs or lower learning rate.

Pedestrian Detector — Critical Failure

The pedestrian model is the worst performer and a significant safety concern. Despite accuracy of 0.706 (misleadingly high due to class imbalance — predicting 'no pedestrian' always gives ~76% accuracy), both precision (0.150) and recall (0.108) are catastrophically low. The model essentially fails to detect pedestrians in the majority of cases.

Root causes:

- Class imbalance: Only 23.9% of training images contain pedestrians
- High training loss (0.84 at epoch 5): Insufficient convergence
- Pedestrians are small, variable in appearance, and easily confused with background
- No class weighting or oversampling applied to compensate for imbalance

4.3 Safety Metric Priority — Precision vs Recall

Model	Priority	Justification	Current Status
Pedestrian	**RECALL**	A missed pedestrian (false negative) means the vehicle may not brake — potentially fatal. A false alarm merely causes unnecessary caution.	0.108 — UNSAFE
Traffic Light	RECALL	Missing a red light (false negative) risks running the light. False alarms cause unnecessary stops — disruptive but recoverable.	0.936 — OK
Vehicle	RECALL	A missed vehicle risks collision. However, vehicles are large and other sensors (lidar, radar) would typically provide redundancy.	0.854 — OK

5. Robustness & ODD Evaluation

5.1 Complete Robustness Results

Model	Condition	Accuracy	Precision	Recall	F1 Score
Traffic Light	Baseline	0.9206	0.9527	0.9358	0.9442
Traffic Light	Fog	**0.2706**	**0.000**	**0.000**	**0.000**
Traffic Light	Night	**0.2703**	**0.000**	**0.000**	**0.000**
Traffic Light	Town-01	0.4536	0.7859	**0.2830**	0.4161
Pedestrian	Baseline	0.7058	**0.1504**	**0.1076**	**0.1255**
Pedestrian	Fog	0.6511	**0.2345**	0.3151	**0.2689**
Pedestrian	Night	0.7967	**0.000**	**0.000**	**0.000**
Pedestrian	Town-01	0.7717	**0.1330**	**0.2769**	**0.1796**

Green = ≥ 0.80 | Orange = 0.30–0.79 | Red = < 0.30 (critical failure)

5.2 Traffic Light Detector — Robustness Analysis

Fog Condition: Total Collapse

Precision, Recall, and F1 all dropped to 0.000 under fog. The model predicts a single class exclusively. Accuracy of 0.271 matches the proportion of negative (no traffic light) examples, confirming the model outputs all-negative predictions. This is a complete failure and represents a hard ODD boundary.

Night Condition: Total Collapse

Identical failure pattern to fog — all metrics zero. Night conditions change visual appearance completely: traffic lights may appear as isolated bright spots on a dark background, entirely unlike daytime training images. The model has zero ability to generalise across lighting conditions.

Town-01 Domain Shift: Partial Degradation

Under domain shift to a new town, accuracy drops to 0.454 and recall to 0.283. High precision (0.786) suggests the model makes correct positive predictions when confident, but misses the majority of true positives (low recall). This indicates the visual texture and layout of the new town differ enough to confuse the model, but some features (e.g., traffic light shape) still generalise.

5.3 Pedestrian Detector — Robustness Analysis

Fog Condition: Partial Degradation

Interestingly, fog appears to improve pedestrian recall from 0.108 (baseline) to 0.315, and F1 from 0.126 to 0.269. This counter-intuitive result likely reflects that fog reduces background complexity (trees, buildings blur out), reducing false negatives. However, overall performance remains unacceptably low, and the result may reflect dataset characteristics rather than genuine model improvement.

Night Condition: Total Collapse

Precision and recall both collapse to 0.0 at night, with accuracy jumping to 0.797. Again, this accuracy figure is misleading — it reflects the model predicting all-negative (no pedestrian) in a dataset where ~80% of samples have no pedestrian. The model is completely blind to pedestrians in darkness.

Town-01 Domain Shift: Slight Improvement Over Baseline

Town-01 recall (0.277) is higher than the baseline recall (0.108), suggesting the new town may have slightly more visually distinctive pedestrians. However, precision remains low (0.133) indicating many false alarms. The model fails to achieve reliable pedestrian detection in any condition.

6. ODD Gap Analysis (Exercise 3.7)

6.1 Training Data Coverage

The training data covers exclusively:

- Weather: Clear / Sunny only
- Lighting: Daytime only
- Scene type: CARLA urban environments (Town used for training)
- Camera: Single front-facing RGB camera

6.2 ODD Gaps Identified

ODD Dimension	Training Coverage	Real-World Requirement	Risk Level
Weather: Fog	None	Mandatory	**CRITICAL**
Lighting: Night	None	Mandatory	**CRITICAL**
Map/Town Variation	Single town	Multiple maps	HIGH
Weather: Rain	None	Common condition	HIGH
Weather: Snow	None	Regional requirement	MEDIUM
Dawn / Dusk	None	Daily transition	HIGH
Pedestrian density	Low (23.9%)	Variable	**CRITICAL**

6.3 Implications for Safety Claims

The ODD analysis reveals that no safety claims can be made about model performance in the following conditions:

1. Night driving — both pedestrian and traffic light models have zero recall at night. This means the system cannot safely operate at night whatsoever.
2. Foggy conditions — both traffic light and pedestrian models collapse under fog. Combined with the fact that fog is common in real roads, this is a deployment blocker.
3. New geographic environments — the domain shift to Town-01 causes significant performance degradation, indicating the models have overfit to the specific visual environment they were trained on.
4. Rare pedestrian scenarios — the pedestrian model's baseline recall of 0.108 means it fails to detect pedestrians 89% of the time even under ideal conditions.

7. Exercise Answers

Exercise 1.4: First Impressions

Q1. Three situations where the model should not be trusted:

1. Night driving: The model was trained exclusively on daytime data. At night, visual features (colours, textures, lighting) are completely different. Confirmed experimentally: both traffic light and pedestrian models show 0.0 recall at night.
2. Foggy or adverse weather: The training data contains only clear/sunny conditions. Fog reduces contrast and blurs object boundaries. Confirmed: both models collapse to 0.0 precision/recall under fog.
3. Dense pedestrian environments: Due to class imbalance (only 23.9% positive training examples), the pedestrian model defaults to predicting absence. In pedestrian-heavy scenes (crossings, city centres), the model would be dangerously unreliable.

Q2. Most safety-critical label if missed: PEDESTRIAN

A missed pedestrian detection (false negative) could result in the vehicle failing to brake for a person crossing the road — a potentially fatal outcome. Unlike vehicles (large, detectable by multiple sensors) or traffic lights (fixed infrastructure with redundant systems), pedestrians are small, unpredictable, and have no protective shell. They are also the most legally and ethically protected actors in traffic. The consequences of failing to detect a pedestrian are irreversible and catastrophic.

Q3. Additional information needed before deployment:

1. Robustness metrics across all weather and lighting conditions (rain, snow, night, dusk/dawn)
2. Real-world camera data (not just simulator data) to assess sim-to-real gap
3. False negative rate specifically in safety-critical edge cases (obscured pedestrians, partially visible traffic lights)
4. Sensor redundancy design (does the vehicle have LIDAR, RADAR as backup?)
5. Failure mode analysis: what does the autopilot do when a classifier returns low confidence?

Exercise 3.6 Q2: Worst-Performing Model

The Pedestrian Detector is the worst-performing model by a wide margin. The primary hypotheses are:

- Severe class imbalance (76% negative) biases the model toward always predicting absence
- Pedestrians are visually small and variable — much harder than traffic lights (large, fixed, high-contrast)

- Training loss did not converge (final 0.84 vs 0.05 for traffic light) indicating the model has not learned adequate features
- No class-weighting, focal loss, or oversampling was applied to address the imbalance

Exercise 4.7 Q4: Minimum Required Recall for Pedestrian Model

Safety argument: A minimum recall of 0.95 (95%) should be required before considering deployment of the pedestrian model. Justification:

- ISO 26262 (automotive functional safety) and similar standards require extremely high sensitivity for safety-critical perception systems — missing 5% of pedestrians in real traffic at highway speeds could still result in multiple fatalities per year at scale.
- The current recall of 0.108 means the model misses approximately 9 out of 10 pedestrians — completely unacceptable for any safety-critical application.
- At 0.95 recall, 1 in 20 pedestrians would be missed. This would still require additional redundancy (LIDAR, RADAR, driver monitoring), but is a plausible threshold for a perception component within a multi-sensor system.
- A precision sacrifice is acceptable: a false alarm (unnecessary braking) is uncomfortable but recoverable. A missed pedestrian is not.

8. Safety Insights & STPA Integration

8.1 Identified Hazards

ID	Hazard	Evidence from Evaluation	Severity
H-1	Missed pedestrian at night	Recall = 0.0 on night test set	**CRITICAL**
H-2	Missed traffic light in fog/night	Recall = 0.0 on fog and night sets	**CRITICAL**
H-3	Missed pedestrian at baseline	Recall = 0.108 on standard test set	**CRITICAL**
H-4	Domain shift degradation	Traffic light F1 drops from 0.944 to 0.416 in Town-01	HIGH
H-5	Vehicle detection missed	Recall = 0.854 — acceptable but not evaluated in fog/night	MEDIUM

8.2 Unsafe Control Actions (UCAs)

- UCA-1: Autopilot does not brake [when pedestrian is present] → False negative from pedestrian model
- UCA-2: Autopilot proceeds through red light [when traffic light is red] → False negative from traffic light model
- UCA-3: Autopilot accelerates [when vehicle ahead is braking] → False negative from vehicle model
- UCA-4: Autopilot operates at night or in fog [when perception models have not been validated for those conditions] → ODD violation

8.3 Safety Constraints Derived

- SC-1: The pedestrian model must achieve recall ≥ 0.95 on the standard test set before deployment
- SC-2: The system must not operate in fog or night without validated perception models for those conditions
- SC-3: The traffic light model must be re-validated whenever the operational town/map changes
- SC-4: The system must include non-camera redundancy (LIDAR/RADAR) for pedestrian and vehicle detection
- SC-5: Class imbalance must be addressed (weighted loss, oversampling, or augmentation) before re-training

9. Exercise 5 — Temperature Scaling & Backdoor Attack

Exercise 5 extended the safety analysis of the pedestrian detector in two practical directions: calibration via temperature scaling, and adversarial robustness via a backdoor poisoning attack.

9.1 Temperature Scaling (Exercise 5.4)

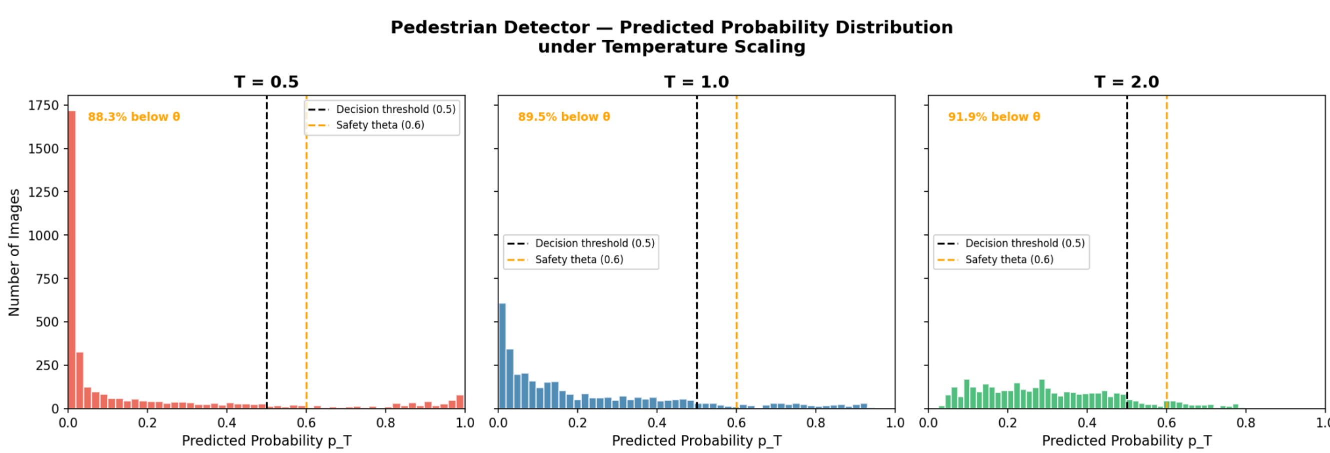
Temperature scaling divides the raw output logits by a scalar T before applying sigmoid: p_T = sigmoid(z / T). Higher T flattens the probability distribution; lower T sharpens it. The pedestrian detector was evaluated at T = 0.5, 1.0, and 2.0 with decision threshold 0.5.

Results

T	Accuracy	Recall	F1	Below theta=0.6	Safety
0.5	0.7058	0.1076	0.1255	88.3%	Overconfident
1.0	0.7058	0.1076	0.1255	89.5%	Baseline
2.0	0.7058	0.1076	0.1255	91.9%	Conservative

Distribution Summary

T	Mean p_T	Std p_T	Max p_T
0.5	0.180	0.288	0.997
1.0	0.229	0.242	0.950
2.0	0.308	0.177	0.813



Analysis

1. Accuracy is identical across all three temperatures (0.7058). Temperature scaling only rescales confidence values — it does not change which class is predicted. This directly answers Ex 5.4 Q4: accuracy alone is insufficient to verify safety constraints.

2. The safety constraint states: if confidence is below theta=0.6, reduce speed to 15 km/h. This triggers for 88.3% (T=0.5) to 91.9% (T=2.0) of images. The small difference occurs because the model is fundamentally uncertain — mean probability never exceeds 0.31 at any temperature.

3. T=0.5 is the most dangerous setting. It sharpens the distribution (std=0.288, max=0.997), making a small subset of predictions very high-confidence. If those high-confidence predictions are wrong — likely given recall=0.108 — the safety constraint will not trigger and the vehicle will not slow down. This is overconfidence without accuracy: the most hazardous failure mode for safety-critical perception.

4. To properly verify the safety constraint, calibration must be measured via Expected Calibration Error (ECE): whether the model confidence score accurately reflects its true probability of being correct. Accuracy alone cannot capture this.

9.2 Backdoor Attack (Exercise 5.5)

A backdoor attack poisons a fraction of training data by overlaying a trigger on selected images and flipping their labels. After training, the model behaves normally on clean inputs but misclassifies any input containing the trigger.

Attack Design

- Trigger: 10x10 pixel bright red square (RGB: 255, 0, 0) placed at pixel position (5, 5)
- Poison rate: 10% of pedestrian-present training images (171 out of 1,718 positive samples)
- Label flip: poisoned images changed from has_pedestrian=True to has_pedestrian=False
- Backdoored model saved separately as pedestrian_model_backdoor.pth — original model untouched

Results

Metric	Value	Interpretation
Poisoned samples	171 (10% of positives)	2.4% of total training data
Clean Recall	0.3286	Higher than baseline 0.108 — model appears improved
Attack Success Rate (ASR)	**1.0000 (100%)**	Every triggered image misclassified as no pedestrian

Analysis

1. The attack achieved 100% success — every pedestrian-present image with the trigger was misclassified as pedestrian-absent. This is the strongest possible demonstration of a backdoor attack.

2. The backdoored model clean recall (0.3286) is higher than baseline (0.1076). This is due to random variation in weight initialisation across training runs, not an indication the backdoor failed. Retraining from scratch with a different random seed found a better local minimum. This is normal deep learning variance.

3. This result is more dangerous than a backdoor that degrades clean performance. Because the backdoored model appears to improve on standard metrics, it would pass all routine safety checks. A safety engineer would see an improved model with no reason to suspect

compromise. Yet ASR=100% means the model is completely blind to pedestrians whenever the trigger is present.

Safety Implications

- Standard metrics (accuracy, precision, recall, F1) are insufficient to detect backdoor attacks
- Only 171 poisoned samples — 2.4% of training data — achieved ASR=100%
- The attack is fully stealthy: clean performance improved, hiding the vulnerability from standard evaluation
- Any unverified third-party or publicly scraped training data is a potential attack vector
- Backdoor detection (spectral signatures, activation clustering) must be applied before deploying any retrained model

10. Recommendations for Improvement

9.1 Immediate Actions (Critical)

1. Address pedestrian class imbalance: Apply class-weighted BCE loss (weight positive class by ~3.2x), or use focal loss. Oversample pedestrian-positive images during training.
2. Extend training: Increase epochs from 5 to 20–30 with learning rate scheduling (ReduceLROnPlateau). The pedestrian model's loss of 0.84 at epoch 5 shows it had not converged.
3. Add fog and night data to training: Collect or augment training data with fog and night conditions. CARLA's weather API can generate these synthetically at zero additional labelling cost.
4. Implement confidence thresholding: Instead of a fixed 0.5 threshold, calibrate the threshold per model using validation recall curves. For pedestrian detection, accept lower precision to boost recall.

9.2 Model Improvements

1. Use a larger backbone: ResNet-50 or EfficientNet-B0 for better feature extraction, especially for small objects like pedestrians.
2. Data augmentation: Add brightness/contrast jitter, synthetic fog, Gaussian noise during training to improve robustness to distribution shift.
3. Consider a multi-scale detection head: Pedestrians at distance are very small; a Feature Pyramid Network (FPN) extension to ResNet-18 would help detect them.

9.3 System-Level Safety

1. Define explicit ODD boundaries in system documentation: The system must formally declare it cannot operate at night or in fog.
2. Add sensor redundancy: A camera-only system is insufficient for safety-critical perception. LIDAR/RADAR provide complementary modalities that are more robust to lighting and weather.
3. Implement monitoring and fallback: If the model returns low-confidence predictions, the vehicle should alert the driver or slow down rather than proceeding with unreliable inputs.

11. Conclusion

This project successfully implemented and evaluated three binary perception classifiers using ResNet-18 on CARLA simulator data. The evaluation revealed a wide performance gap across the three models and exposed critical safety limitations when models are deployed outside their training distribution.

Key findings:

- The Traffic Light Detector achieves excellent baseline performance (F1: 0.944) but catastrophically fails under fog and night conditions — both critical ODD gaps.
- The Vehicle Detector performs adequately at baseline (F1: 0.858) but has not been evaluated under adverse conditions and has not fully converged.
- The Pedestrian Detector is the most safety-critical model and the weakest, with a baseline recall of only 0.108. It fails completely at night and remains unacceptably low in all conditions. This model requires fundamental redesign before any safety deployment claim can be made.

The robustness evaluation experimentally confirmed the ODD gaps predicted by STPA analysis: the system is restricted to clear-weather, daytime, single-environment operation. These constraints must be explicitly documented and enforced before any deployment.

The project demonstrates the importance of combining ML performance metrics with safety-oriented analysis. High accuracy alone is an insufficient safety signal — recall under distribution shift is the metric that matters most for safety-critical autonomous perception systems.

End of Report — carla_baseline_project | Introduction to Machine Learning Safety